Supplementary Material for "FaceGuard: Continuous Face Authentication With a Homography-Parallax Liveness Cue for Commodity Webcams"

First A. Author, *Student Member, IEEE,* and Second B. Author

This document supplements the main article. References to sections, figures, tables and equations without the prefix S, such as Section III-B or Table I, point to the main article; the references at the end are numbered independently of those of the main article.

S1. System Details

Comparison with alternatives. Table S1 lists the functions that FaceGuard gives the face camera next to an idle-timeout lock and IR-based Windows Hello face sign-in [1]. FaceGuard deliberately does not unlock the operating system (OS), so that fooling its RGB face matcher can never grant access to the OS (Section III-A).

Tracking. Detections are linked to tracks greedily by intersection over union (IoU ≥ 0.25). If no box overlaps enough, a detection whose center lies within 0.6 face widths of a track and whose size differs by less than a factor of 1.5 continues that track, which handles fast motion at low frame rates; a track ends after five missed frames or 1 s. Each track is re-embedded every 0.35 s (0.2 s in the background).

Quality gate. For enrollment the gate requires a face at least 80 pixels wide, a Laplacian variance of at least 35 (gray levels squared, on the 112×112 crop), a mean brightness between 45 and 200 gray levels with at most 12% over-exposed pixels, a detector score of at least 0.8, $|\text{yaw}| \leq 40°$, $|\text{pitch}| \leq 35°$ and the whole face inside the frame. Passive decisions (template updates and intruder locks) use 60 pixels, a Laplacian variance of 25 and $|\text{yaw}|, |\text{pitch}| \leq 30°$. These limits are deliberately conservative relative to the degradation results of Section S2, where recognition degrades noticeably below about 24 pixels and under over-exposure. The shoulder-surfer rule does not require the gate, because only a notification follows.

Presence guard. Fig. S1 shows the states of the presence guard and Algorithm S1 its update rule, which runs four times per second (Section III-F).

Implementation. OpenCV runs the YuNet (`face_detection_yunet_2023mar`, 0.23 MB) and SFace (`face_recognition_sface_2021dec`, 38.7 MB) models in the Open Neural Network Exchange (ONNX) format, and the MediaPipe Face Landmarker (`face_landmarker.task`, 3.8 MB) supplies 478 points and 52 blendshape scores. Template files are written atomically with a backup copy. The application offers a live view with landmark, mesh and

TABLE S1
Functions of the Face Camera Compared

Function	Timeout	Hello [1]	FaceGuard
Unlock the OS with the face	–	yes	no (by design)
Lock when the user walks away	delayed	–[a]	yes
Lock and photograph a stranger	–	–[a]	yes
Watch the locked PC	–	–[a]	yes
Warn about an onlooker	–	–[a]	yes
Recognize several people in view	–	–[a]	yes
Liveness	–	IR imaging	challenges
Camera required	none	IR camera	RGB webcam

Timeout: idle-timeout lock; Hello: Windows Hello face sign-in. [a]Not a function of face sign-in, which authenticates the user only when the session is unlocked.

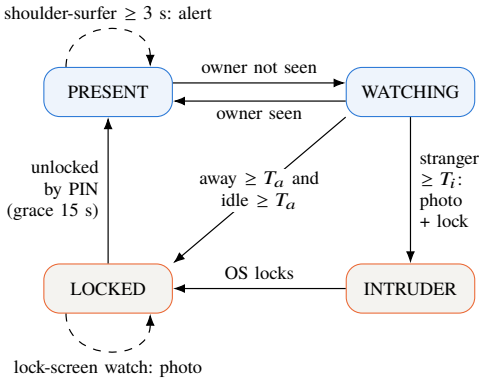


Fig. S1. Presence-guard states (simplified; PAUSED, NO_CAMERA and NO_OWNER omitted). Solid arrows change state, dashed arrows raise alerts.

Algorithm S1 Presence-guard update (4 frames per second)

Require: labeled tracked faces, idle time $I(t)$, session state
Ensure: possibly a lock, a photograph or a notification
if the session is locked **then**
  if within W min of locking and a stranger is confirmed **then** photograph it
  return
end if
if an authorized face is visible **then** $t_a \leftarrow t$
if $\min(t - t_a, I(t)) \geq T_a$ **then** lock the PC (warning at $T_a - 8$ s)
else if a quality-checked stranger is seen for T_i, $t - t_a \geq 3$ s, outside the grace period, < 3 locks in 10 min **then**
  photograph, notify and lock the PC
else if an authorized face and a stranger ≥ 40 px are seen for 3 s **then**
  notify (at most once per minute)
end if

head-pose overlays, guided enrollment, face unlock, people management, the presence guard, an intruder gallery and a "Model Lab" that shows leave-one-out accuracy, score histograms and an embedding map of the user's own gallery (Fig. S2). All functions are also available from the command line. The code base has about ten thousand lines of Python and 110 automated tests.

S2. Recognition

Data details. The 13,233 LFW photographs show 5,749 individuals. YuNet found a face in every image; in the 2,777 images with several faces, the one nearest to the image center was used. In the median, faces are 93 pixels wide and turned by 7.2°. The head-pose agreement reported in Section III-C was measured on the 2,998 of 3,000 randomly sampled LFW images for which MediaPipe returned a mesh.

Identification and robustness protocols. Identification involves the 158 people with ten or more images (4,324 images in total) and g enrollment images each, averaged over five repetitions. For the open-set test, half of these people (79) are enrolled with five images, and the probes are their remaining images (1,781 on average) together with every image of the others (11,057). Robustness is measured on the 1,200 pairs of View-2 splits 1 and 2 with only the probe degraded; in addition, every degraded probe is compared with 1,000 clean images of other people, giving 1,199,435 impostor comparisons per condition.

Verification. Fig. S3 shows the ROC curves of the SFace variants of Table I, Table S2 the operating points of the three security levels on single LFW image pairs, and Fig. S4 the underlying score distributions.

Identification. Closed-set identification is already 99.30% correct with a single enrollment image and levels off near 99.9% with two or more. Classifiers trained on the embeddings bring no meaningful gain: with five enrollment images a linear support vector machine reaches 99.93%, logistic regression 99.92% and k-nearest neighbors 99.92%, against 99.92% for the top-3 rule, and unlike the similarity rules they must be retrained for every new person and cannot reject strangers by design. The open-set test separates the rules more clearly (Fig. S5, Table S3). At the balanced threshold, the top-3 rule falsely accepts 1.22% of unseen people, about half the rate of the nearest-neighbor rule (2.36%), while identifying 99.65% of enrolled probes; at the strict level its false-alarm rate is 0.08%. The centroid rule attains the highest detection-and-identification rate (DIR) at low false-alarm rates but depends on a representative enrollment, whereas the top-3 mean copes better with atypical samples and is therefore the default. False alarms are more frequent here than in Table S2 because each probe is compared with every enrolled person, which is why the guard requires unknown faces to persist and face unlock requires a live, smoothed match.

Robustness to degradations. Fig. S6 shows verification with degraded probes at the balanced threshold, where clean probes give 97.00% TAR. Darkness hardly matters (88.17% TAR at 15% brightness), whereas over-exposure does (25.17% at three times the brightness), which motivates the asymmetric exposure limits of the quality gate. Recognition holds for faces down to about 24 pixels in width (85.83% TAR), while detection breaks down below roughly 17 pixels, where only

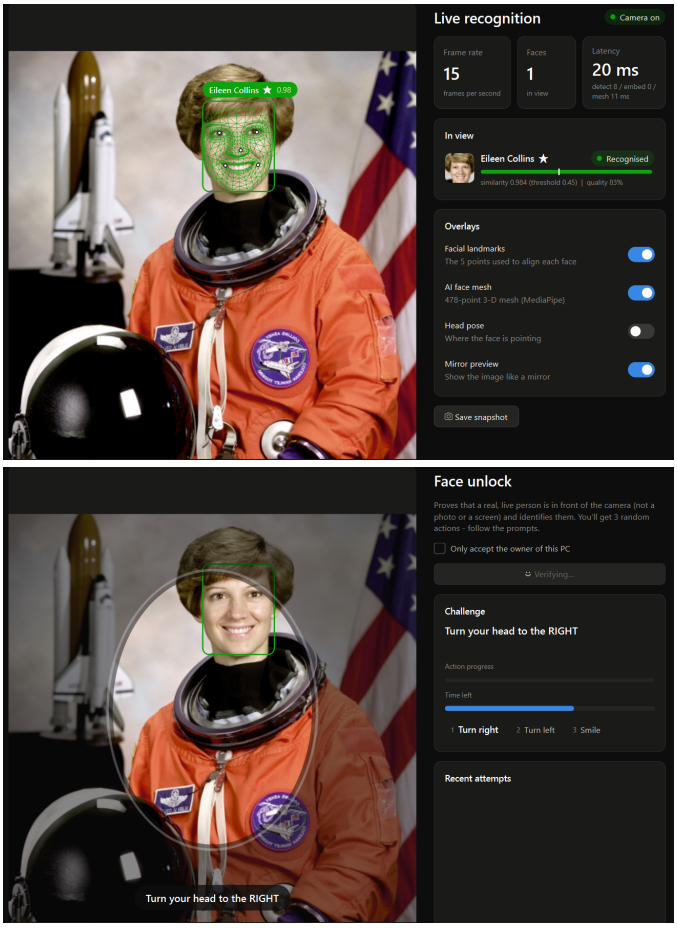


Fig. S2. FaceGuard user interface (navigation bar cropped). Top: live recognition with the 478-point face mesh, the identity label with its similarity meter and threshold marker, and the per-stage latency. Bottom: a face-unlock session prompting a head turn, for which a still photograph cannot produce the required parallax. The camera input is replaced here by a public-domain NASA portrait.

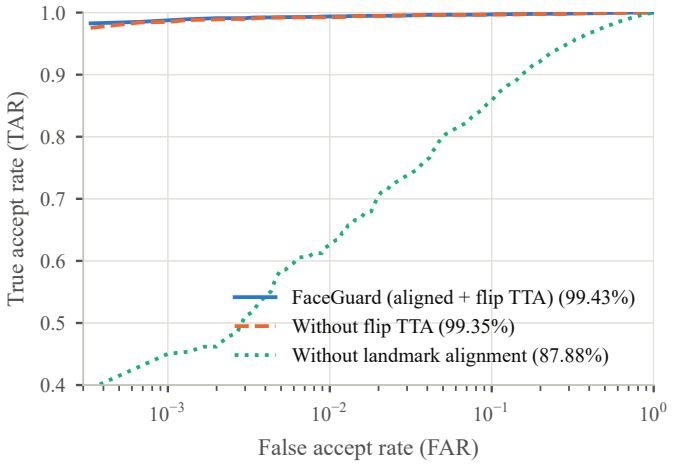


Fig. S3. ROC curves of the SFace pipeline on LFW View 2 (accuracy in brackets). Landmark alignment is essential; flip test-time augmentation (TTA) adds a small gain.

TABLE S2
OPERATING POINTS ON LFW IMAGE PAIRS (52,904 GENUINE AND 1,000,000 IMPOSTOR PAIRS)

Security level	τ	FAR (%)	FRR (%)
Relaxed	0.38	0.0628	0.88
Balanced (default)	0.45	0.0054	2.54
Strict	0.52	0.0004	7.79
Windows Hello requirement [1]	–	< 0.001	< 5

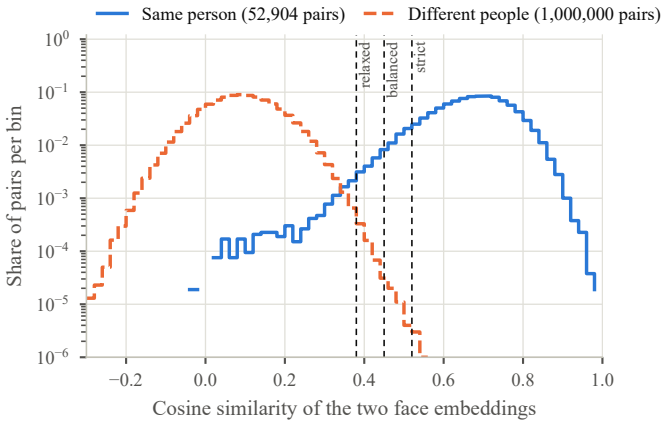


Fig. S4. Similarity distributions of 52,904 genuine and 1,000,000 impostor LFW pairs (log scale) with the three security thresholds.

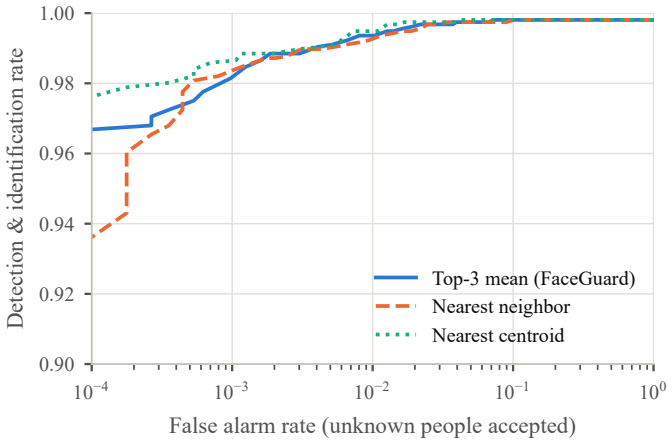


Fig. S5. Open-set identification on LFW: 79 enrolled people with five images each, probed by their other images and by 11,057 images of unseen people.

TABLE S3
OPEN-SET IDENTIFICATION ON LFW: DIR (%) AT FIXED FALSE-ALARM RATE AND DIR / FALSE-ALARM RATE (%) AT THE SECURITY LEVELS

Rule	DIR at FAR 1%	DIR at FAR 0.1%	Relaxed	Balanced	Strict
Nearest neighb.	99.57	98.87	99.84/16.06	99.73/2.36	99.15/0.35
Top-3 (FaceGuard)	99.61	99.01	99.83/4.67	99.65/1.22	98.59/0.08
Centroid	99.69	99.36	99.87/9.56	99.80/1.99	99.41/0.44

15.8% of faces are found. Gaussian blur of 4 pixels, JPEG quality 4 and noise with a standard deviation of 32 gray levels reduce the TAR to 46.83%, 18.33% and 42.17%, respectively. A mask-like occluder over the lower face keeps 87.33% TAR and sunglasses 71.67%, with AUCs of 0.9993 and 0.9976, so occlusion harms a fixed threshold more than it harms the ranking of scores.

On the 1,199,435 impostor comparisons per condition, clean probes produced 36 false accepts (FAR 0.0030%). Severe degradations lowered the FAR (at most 0.0017% for blur of 4 or 6 pixels, three-fold over-exposure, down-scaling to 12%, JPEG quality 4 or noise of 32 gray levels), whereas mild ones left it of the same order; the highest value, 0.0047% (56 false accepts) for down-scaling by a factor of 0.5, remains below the pairwise FAR of the balanced level in Table S2. Degraded probes are therefore mostly rejected rather than confused with other people, but mild down-scaling does not by itself make the system safer.

S3. LIVENESS

Simulation details. The simulated head faced the camera, turned at a random speed and, when the progress bar stalled, turned further or blinked harder. Each print configuration was attacked 1,000 times and the two replay attacks were run 800 and 1,500 times; live users were simulated in 2,000 sessions for the default protocol and in 1,000 sessions per ablation.

Sensitivity to the behavioral model. Table S4 varies the behavioral model of simulated live users. The BPCER is unchanged for median reaction delays of 0.5 and 1.0 s (1.1% and 1.4%) and stays low for larger turns (1.2% at 38°) or occasional confusion (2.4% at 10%); it rises to 2.6% for very slow users (1.4 s), 2.1% for timid turns of 22° and 3.2% when left and right are confused in 20% of prompts. Landmark noise and stricter turn thresholds also raise rejections (Fig. S7(a)); with 2-pixel noise the BPCER of the default protocol is 1.8%. A session lasted 4.1 s on average (Fig. S7(b)).

Pose jitter. Under sensor noise of 4 gray levels, the yaw estimate of a still face fluctuates by 0.66° on average (95th percentile 1.62°), far below the 18° turn threshold.

S4. PRESENCE GUARD AND RUNTIME

Guard simulation. Table S5 gives the full results of the guard simulation of Section V-G. In the owner-at-the-desk scenario the owner was labeled unknown in 0.36% of frames, but an authorized face was present in every frame (longest period without one: 0.0 s, against T_a = 30 s), so the frames labeled unknown are frames in which the largest face in view was another person in the photograph. Re-running the production engine on the streamed photographs traced the 279 shoulder-surfer alerts to the 224 of 1578 images (14.2%) in which a second, unknown person at least 40 pixels wide is visible; cues such as gaze direction or distance could make the rule more selective.

Classifying each trial's stranger with the production engine explains every miss of the intruder and onlooker trials: the quality gate rejected the photograph in 7 and 13 trials and the score fell in the uncertain band in 2 and 2 trials, respectively, while no stranger was ever accepted as the owner. The replacement trials, which reuse the same strangers, missed 2 more.

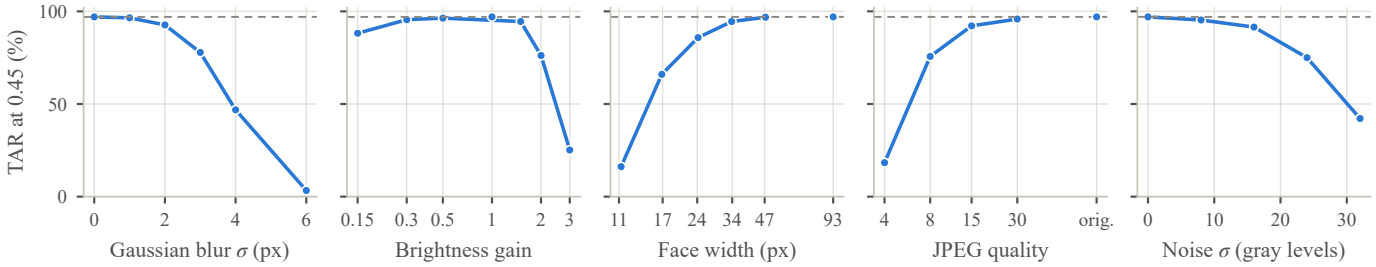


Fig. S6. Robustness of verification at the balanced threshold ($\tau = 0.45$) when the probe is degraded (1,200 LFW pairs, enrollment image clean). The dashed line is the clean-probe TAR.

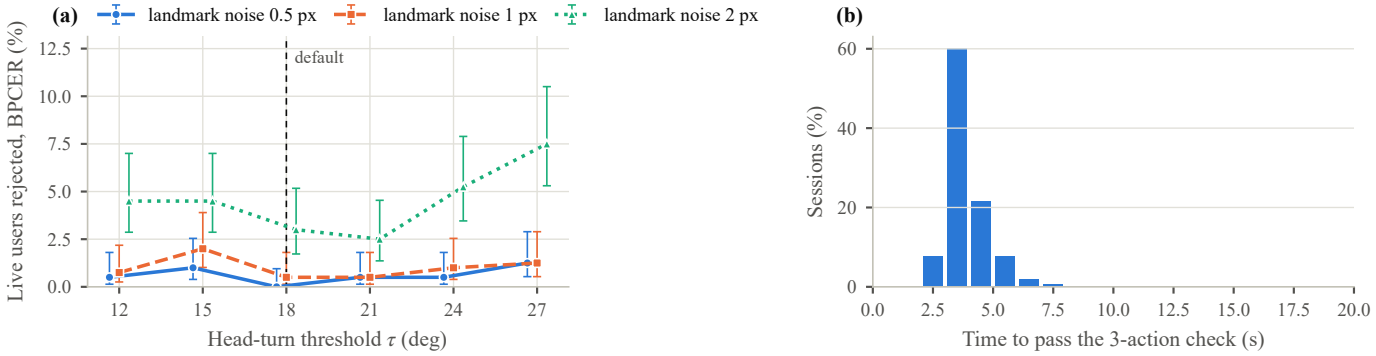


Fig. S7. Live-user rejection. (a) BPCER against the head-turn threshold for three levels of landmark noise (turn-only protocol, 400 sessions per point, 95% Wilson intervals). (b) Time needed to pass the default three-action check (2,000 sessions).

TABLE S4
SENSITIVITY OF THE BPCER (%) TO THE BEHAVIORAL MODEL OF LIVE USERS (DEFAULT VALUES IN BOLD)

Median reaction delay (s)	0.5	**0.7**	1.0	1.4
BPCER	1.1	**1.4**	1.4	2.6
Mean turn amplitude (°)	22	**30**	38	
BPCER	2.1	**1.4**	1.2	
Left/right confusion (%)	0	**5**	10	20
BPCER	0.9	**1.4**	2.4	3.2

1,000 simulated users per non-default setting (2,000 for the default).

TABLE S5
PRESENCE GUARD ON SIMULATED WEBCAM STREAMS (PRODUCTION ENGINE AND THRESHOLDS, 4 FRAMES PER SECOND)

Owner at the desk: 40 owners, 10.0 h, 144,000 frames	
frames: owner / uncertain / unknown (%)	99.64 / 0.00 / 0.36
false intruder locks per hour	0.00
false walk-away locks per hour, no typing	0.00
longest time without recognition (s)	0.0
shoulder-surfer alerts per hour (see text)	27.9
Stranger sits down 5 s after the owner left (300 trials)	
locked within 20 s (%)	97.0
time to lock: median / mean / 95th pct. (s)	2.75 / 2.87 / 2.75
Stranger replaces the owner without a gap (300 trials)	
locked within 20 s (%)	96.3
time to lock: median / mean / 95th pct. (s)	2.75 / 2.96 / 3.25
Shoulder-surfer behind the owner (150 trials)	
alerted within 20 s (%)	90.0
time to alert: median / 95th pct. (s)	3.25 / 3.25

Latencies are quantized to the 0.25 s frame interval, so the median and the 95th percentile can coincide.

Runtime. Fig. S8 breaks down the latency at the default 640 × 480 resolution: detection takes 12.1 ms, alignment 0.06 ms and one embedding 9.3 ms, or 18.5 ms with mirroring. A frame of the complete pipeline costs 12.5 ms on average, 21.0 ms if every frame is re-embedded and 23.5 ms with the face mesh. The live view ran at 29.0 frames per second with a total CPU load of 19.4%, while the background guard at four frames per second used 5.1% (82% of one core). Loading the recognition models takes 101 ms and the mesh model 64 ms. With flip TTA and default threading, an SFace embedding of an aligned crop takes 18.4 ms, an ArcFace MobileFaceNet embedding 11.4 ms and an ArcFace ResNet-50 embedding 96.1 ms. Runtimes are means of 200 runs after 20 warm-up runs on an otherwise idle machine; the CPU load of the live view was measured with a synthetic 30-frame-per-second image stream instead of a physical camera.

S5. REPRODUCIBILITY

Software. The software is written in Python 3.12 with OpenCV 5.0, NumPy 2.5, MediaPipe 1.0.1, scikit-learn 1.9.1 and PySide6 6.11; the ArcFace baselines use onnxruntime 1.30. Every model is checked against a pinned SHA-256 hash after download.

Seeds. Every randomized step uses a fixed seed: 2026 for the impostor pairs, the identification repetitions, the calibration/evaluation split of the face meshes, the MPIIFaceGaze sample and the demographic impostor pairs; 0 for the mesh selection; 17 for the robustness impostor gallery; 70 and 7 for the two guard scenarios; and one recorded seed per Monte-Carlo configuration.

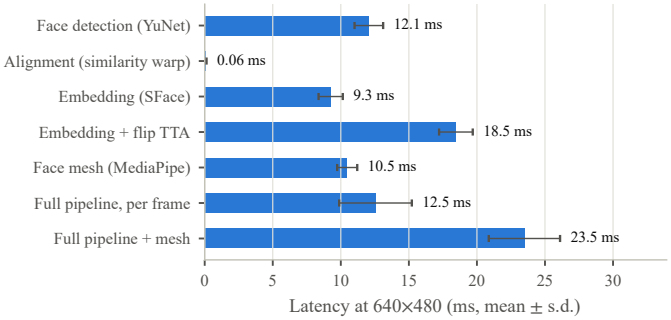


Fig. S8. Latency of each stage on the laptop CPU at 640×480 (200 runs). The full pipeline re-embeds each tracked face every 0.35 s.

Code and data. The experiment scripts regenerate every number, table and figure of the main article from the public datasets, which are downloaded separately under their own licenses (LFW, MPIIFaceGaze, LFW attribute labels) together with the pretrained models. The raw logs of the real-world pilot are not released because they contain per-frame facial landmarks of a participant; their anonymous summary, from which Table VII of the main article is computed, is included.

References

[1] Microsoft, "Windows Hello biometric requirements," Microsoft Learn. [Online]. Available: https://learn.microsoft.com/en-us/windows-hardware/design/device-experiences/windows-hello-biometric-requirements, accessed: Sep. 28, 2026.